

$$\begin{aligned}\mathcal{F}[e^{-at}u(t)] &= \int_0^{\infty} e^{-at} e^{-j2\pi ft} dt \\ &= \int_0^{\infty} e^{-(a+j2\pi f)t} dt \\ &= \frac{1}{a+j2\pi f}\end{aligned}$$